

Problem 1

Let H be a graph with 6 vertices and 9 edges, with the vertex and edge sets defined as follows:

- $V(H) = \{1, 2, 3, 4, 5, 6\}$
- $E(H) = \{\{1, 3\}, \{1, 4\}, \{2, 3\}, \{2, 4\}, \{3, 5\}, \{3, 6\}, \{4, 5\}, \{4, 6\}, \{5, 6\}\}$.

Prove that for every $k > 2$, the optimal graphon W of the following optimisation problem

$$\min_W t(H, W) \quad \text{subject to} \quad t(K_2, W) = 1 - \frac{1}{k}$$

is neither a balanced k -partite graphon nor a constant graphon.

For the specific case when $k = 5$, prove that the optimal value is $\frac{24349}{187500} \cong 0.12986133\dots$ with the optimal graphon structure given by

$$W(x, y) = \begin{cases} 1 & \text{if } 0.1 \leq |x - y| \leq 0.9, \\ 0 & \text{otherwise.} \end{cases}$$

Note

The statement is known to be true for $k = 3, 4$, and 5 with numerical optimisation. However, generalising the counterexample to all $k > 2$ is non-trivial (as found numerical optimums do not share the same structure for different k) and the provided example for $k = 5$ fails to be optimal for $k = 3$ and $k = 4$, if we replace $0.1, 0.9$ with $1/3, 2/3$ and $0.25, 0.75$ respectively.

Problem 2

Consider all graphs with at most 5 vertices that satisfy the following conditions: (1) no isolated vertices, (2) no isolated edges, and (3) that are not bipartite.

For each graph H in this collection, consider the following curve defined by the optimisation problem:

$$f_H(x) = \min_W t(H, W) \quad \text{subject to} \quad t(K_2, W) = x$$

where H is a fixed graph from the collection and $x \in [0, 1]$.

Prove that for every graph H in the collection:

- When H is K_3 , K_4 , K_4 with a pendant edge, a graph obtained by gluing one edge of K_3 and K_4 , or K_5 , the function f_H is piecewise concave, with each piece on the interval $[1 - \frac{1}{k}, 1 - \frac{1}{k+1}]$ for $k \geq 2$.
- When H is not one of the above graphs, the function f_H is piecewise convex, with each piece on the interval $[1 - \frac{1}{k}, 1 - \frac{1}{k+1}]$ for $k \geq 2$.

Moreover, extend this classification to every graph H with more than 5 vertices.

Note

For cliques, it was proven that the curve defined by the optimisation problem is piecewise concave, with each piece on the interval $[1 - \frac{1}{k}, 1 - \frac{1}{k+1}]$ for $k \geq 2$. However, for C_5 , it is conjectured that the curve defined by the optimisation problem is piecewise convex, with each piece on the interval $[1 - \frac{1}{k}, 1 - \frac{1}{k+1}]$ for $k \geq 2$, and there exists a construction that achieves the piecewise convex curve.